Navigation and Positioning Systems in Autonomous Vehicles

Group 3

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Fundementals of autonomous vehicle navigation

NAVIGATION PIPELINE

System Architecture Overview

01 Perception
Environmental Awareness

Detecting and identifying objects using LiDAR, Radar, and Cameras. Sensor fusion creates a unified 360° world model.

LiDAR | Radar | Camera

02 Localization
Precise Positioning

Achieving centimeter-level accuracy through HD Maps, Odometry, and SLAM. Overcoming GPS limitations for lane-level precision.

HD Maps | IMU | SLAM

03 Planning
Decision Intelligence

Three-stage decision architecture: Global Planning, Behavioral Planning, and Local Trajectory Planning using A* and RRT.

A* | RRT | FSM

04 Control
Motion Execution

Converting trajectories into physical commands through actuation and feedback loops using PID and MPC controllers.

PID | MPC | Feedback

LAYER 01

Perception

What is around me?

The perception layer mimics human senses using electronic sensors to gain a 360° view of the environment, detecting and identifying objects with precision.

PERCEPTION LAYER

Sensor Suite: The Eyes of Autonomous Vehicles

LiDAR

LightDetection and Ranging

Uses laser pulses to create high-resolution 3D point clouds of the environment. Provides precise distance measurements and object geometry.

360°	+-2cm	100m+
Coverage	Accuracy	Range

Radar

RadioDetection and Ranging

Excellent for detecting speed and distance of objects. Performs reliably in adverse weather conditions like fog and rain.

All	Doppler	200m+
Weather	Velocity	Range

Cameras

ComputerVision Systems

Essential for reading traffic signs, lane markings, and traffic light colors. Provides rich semantic information.

Semantic	Color
Understanding	Recognition

Why Multiple Sensors?

No single sensor is perfect. Each has unique strengths and weaknesses that complement each other:

- LiDAR: Poor in heavy rain/fog
- Radar: Lower resolution
- Cameras: Struggle in low light

Sensor Fusion: Creating a Unified World Model

The Fusion Challenge

Sensor fusion algorithms combine data from all sources to create a "World Model" that is more accurate and robust than any individual sensor can provide alone.

Real-World Fusion Strategy: LiDAR provides accurate 3D structure, Radar adds velocity and weather resilience,
while Cameras contribute semantic understanding. The fusion algorithm weights each
sensor's contribution based on confidence and environmental conditions.

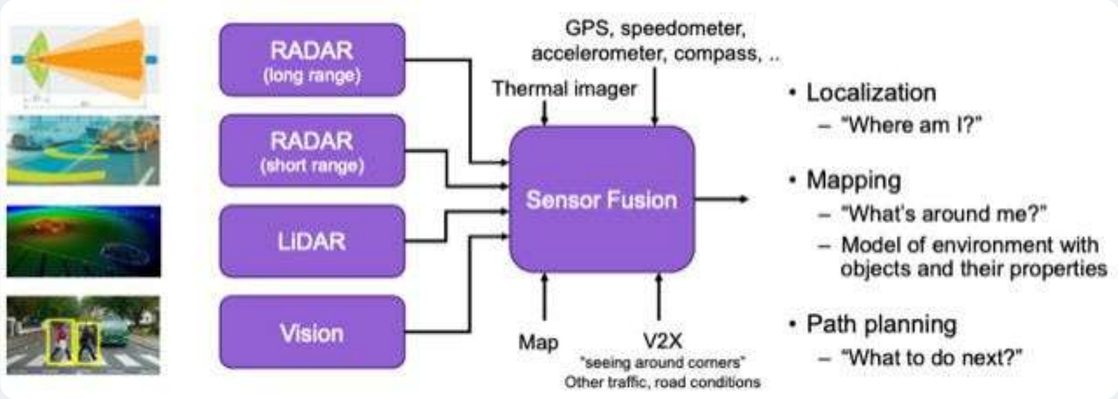
Kalman Filters

The most widely used fusion algorithm for state estimation. It optimally combines predictions with measurements to estimate object states (position, velocity).

Prediction Step
Uses motion model to predict next state

Update Step
Combines prediction with sensor measurement

Sensor Fusion Architecture



Multi-sensor data integration for localization, mapping, and path planning

Fusion Approaches

- **L** — Low-Level Fusion
  Rawdatafusionbefore feature extraction
- **M** — Mid-Level Fusion
  Feature-levelfusionfrom each sensor
- **H** — High-Level Fusion
  Decision-levelfusionof object tracks

LOCALIZATION LAYER

Achieving Centimeter-Level Precision

HD Maps

High-DefinitionPre-Mapped Environments

Pre-mapped, highly detailed 3D maps that include lane widths, curb heights, traffic signal locations, and road geometry with centimeter-level accuracy.

Lane Geometry
Exactlaneboundaries & widths

Semantic Info
Signs,signals,road markings

Localization
Unique feature matching

~10cm
Positioning accuracy

HD Map Visualization

High-definition map with lane-level detail

Odometry

Measuring wheel rotations and using IMUs to track movement from a known starting point.

IMU | Encoders

SLAM

Used when maps aren't available; builds a map while tracking location within it.

Mapping | Localization

Accuracy Comparison

Standard GPS	3-5 meters
RTK-GPS	10-30 cm
HD Maps + Fusion	~10 cm

SYSTEM INTEGRATION

The Complete Navigation Pipeline

Core Navigation Pipeline Summary

01 Perception
Object Detection

Primary Task
Environmental awareness

Key Technology
LiDAR, Radar, CNNs

02 Localization
Positioning

Primary Task
Centimeter accuracy

Key Technology
GNSS, IMU, HD Maps

03 Planning
Decision Making

Primary Task
Path & behavior

Key Technology
A*, Dijkstra, FSM

04 Control
Execution

Primary Task
Motion execution

Key Technology
PID, MPC

Pipeline Integration

The four layers operate as an integrated system: Perception provides environmental data, Localization determines precise position, Planning decides actions, and Control executes them. Each layer depends on the previous, creating a robust Sense-Plan-Act loop.

Robotics Connection

This architecture extends beyond autonomous vehicles to robotics applications:

- ✓ Robot arms use similar control loops
- ✓ Solar trackers employ PID control
- ✓ Mobile robots leverage SLAM

GNSS in Autonomous Vehicles

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01

GNSS OVERVIEW

- Satellite-based positioning system
- Provides position, velocity, and timing
- Key Components of GNSS:

  Space Segment

  Control Segment

  User Segment

How GNSS Works & Trilateration

02

- Satellites send time and position signals
- Receiver measures signal delay
- Distance = speed of light × time
- Each satellite gives a distance

Trilateration:

- Uses distances from multiple satellites
- 3 satellites → 2D position
- 4 satellites → 3D position (including altitude)
- More satellites improve accuracy and reliability
- Reduces errors using redundancy

Applications in Autonomous Vehicles

Localization

- Determines vehicle position on maps
- Helps correct sensor errors

Path Planning

- Supports route planning
- Uses real-time positioning data
- Optimizes driving paths

Tracking

- Enables real-time vehicle tracking
- Monitors movement and performance

Integration with Sensors

- Works with INS, LiDAR, and cameras
- Improves accuracy and reliability
- Ensures continuous positioning

03

04

Limitations of GNSS

- Signal Blockage
  - Urban areas with tall buildings
  - Tunnels and underground roads
- Multipath Effect
- Atmospheric Effects
  - The ionosphere
  - The troposphere
- Limited Accuracy

Improving GNSS Accuracy

DGNSS
Differential GNSS enhances positioning accuracy significantly.

INS Support
Inertial Navigation Systems maintain positioning during loss.

Sensor Fusion
GNSS is combined with INS LiDAR and cameras

Kalman Filter
combine data from GNSS and other sensors reduce noise

RTK

Real-Time Kinematic

Multi-Constellation GNSS
Using multiple satellite systems

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احمد محمد سالم

Inertial Navigation Systems (INS)

01

Why INS is Essential in Autonomous Navigation?

- Evolution of navigation systems in autonomous vehicles
- Role of GNSS and its limitations (signal loss – interference)
- Need for alternative systems
- Definition and importance of INS

HOW INS WORKS:
PRINCIPLES & CORE COMPONENTS

02

INS principle

(acceleration → velocity → position)

01 IMU

02 Accelerometer

03 Gyroscope

System components

Overcoming INS Challenges: Errors & Integration

Error sources (Drift – Noise – Bias)

Error accumulation problem

Integration of INS with GNSS (Kalman Filter)

03

LiDAR-Based Localization

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INTRODUCTION & IMPORTANCE

LiDAR in Autonomous Vehicles

- LiDAR stands for Light Detection and Ranging
- Autonomous vehicles need accurate navigation & positioning
- It works by sending laser pulses and measuring the time taken for the reflected signals to return.

Why LiDAR is Important

- Localization (vehicle position)
- Mapping the environment (3D View)
- Object detection
- Safe navigation

3D MAPPING & LOCALIZATION

02

3D Point Cloud Mapping

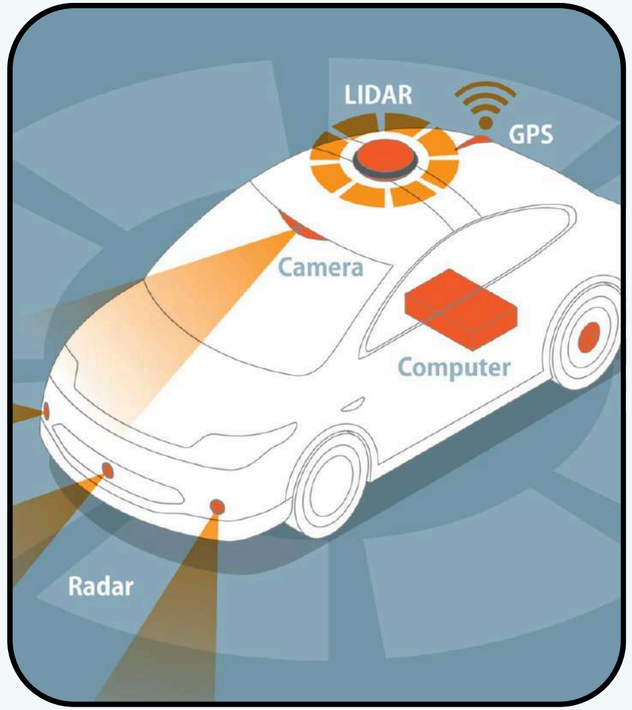
- LiDAR generates a 3D point cloud
- Represents:
  - roads
  - buildings
  - vehicles
  - obstacles

- When all these points are combined, they form a 3D model of the road environment.

Localization

- Compare current scan with:
  - stored maps
  - previous scans
- Helps vehicle know its exact position.
- This is especially useful in areas where GPS is weak or unavailable.

Object Detection



Object Detection using LiDAR

LiDAR detects:

- cars
- pedestrians
- cyclists
- road barriers

Why Important

Helps in:

- braking
- lane changing
- obstacle avoidance

Static vs Dynamic Objects

- Static → buildings, poles, walls (for localization)
- Dynamic → cars, pedestrians (for safety navigation)

03

SLAM Integration

04

What is SLAM?

Simultaneous Localization and Mapping

- Build map + determine position at the same time

Why Important

- the vehicle cannot always depend only on prebuilt maps.
- In some cases, it needs to update its own map while driving.
- Useful for navigation in unknown environments

SLAM Steps (Workflow)

- Scan environment
- Extract features
- Estimate movement
- Match scans
- Update map

Challenges & Future of LiDAR

Challenges

- Large data size
- High computation
- Weather sensitivity
- High cost

Even with these challenges, LiDAR is still one of the strongest technologies for accurate-autonomous positioning.

Future Developments

- Smaller and cheaper LiDAR sensors
- Better real-time processing
- Improved SLAM systems
- Stronger AI integration
- More accurate positioning

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01

INTRODUCTION

OVERVIEW

- Autonomous vehicles rely on advanced navigation and positioning systems.
- Accurate localization is essential for safe and efficient driving.
- Visual SLAM and camera-based positioning provide real-time environmental awareness.
- These technologies enable vehicles to operate even when GPS signals are unreliable.

OBJECTIVES

- Understand Visual SLAM and its role in autonomous navigation.
- Explore camera-based positioning techniques.
- Highlight real-world applications and industry implementations.

FUNDAMENTALS OF VISUAL SLAM

02

WHAT IS VISUAL SLAM?

- Visual SLAM stands for Visual Simultaneous Localization and Mapping.
- It allows a vehicle to build a map while determining its location within that map using cameras.

KEY COMPONENTS

- Feature Detection and Matching
- Pose Estimation
- Mapping and Localization
- Loop Closure
- Optimization

ADVANTAGES

01 Cost-effective compared to LiDAR.

02 Provides rich visual and semantic information.

03 Suitable for urban and indoor environments.

03

CAMERA-BASED POSITIONING TECHNIQUES

MAJOR TECHNIQUES

01 Landmark-Based Localization: Uses road signs, lane markings, and buildings as reference points.

02 Map-Based Localization: Matches real-time images with high-definition maps.

03 Visual-Inertial Odometry (VIO): Combines cameras with IMU sensors for improved accuracy.

04 Perspective-n-Point (PnP): Estimates camera pose from 3D-to-2D correspondences.

BENEFITS

01 High precision and reliability.

02 Low hardware cost.

03 Essential for GPS-denied environments.

04

VISUAL ODOMETRY AND SENSOR FUSION

A. Visual Odometry

- Estimates vehicle motion using sequential images.
- Types:
- Monocular Visual Odometry
- Stereo Visual Odometry
- RGB-D Visual Odometry

B. Sensor Fusion

- Combines multiple sensors to enhance(accuracy and robustness.
- Common Integrations:
  - Camera + LiDAR
  - Camera + Radar
  - Camera + IMU

E. Filtering Techniques

- Extended Kalman Filter (EKF)
- Unscented Kalman Filter (UKF)
- Particle Filters

INDUSTRY APPLICATIONS

REAL-WORLD IMPLEMENTATIONS

- **TESLA**
  Vision-based autonomous driving using camera-centric systems.
- **WAYMO**
  Combines cameras with LiDAR and radar for precise localization.
- **NVIDIA**
  Provides AI platforms such as NVIDIA DRIVE.
- **MOBILEYE**
  Specializes in camera-based driver assistance systems.
- **APPLE**
  Develops advanced perception technologies for autonomous mobility.

CHALLENGES, FUTURE TRENDS, AND CONCLUSION

REAL-WORLD IMPLEMENTATIONS

CHALLENGES

- Lighting variations and weather conditions.
- Dynamic environments and occlusions.
- High computational requirements.
- Scale drift in monocular systems.

FUTURE TRENDS

- Deep Learning-Based SLAM.
- Edge and Cloud Computing.
- Collaborative and Cloud SLAM.
- Integration with 5G and V2X technologies.

CONCLUSION

- Visual SLAM and camera-based positioning are essential for autonomous vehicles.
- They enable accurate localization, mapping, and navigation.
- Continued advancements will shape the future of intelligent transportation.

مصطفي حسن جاب الله

01

AUTONOMOUS VEHICLES & THE ROLE OF NAVIGATION SYSTEMS

- Autonomous vehicles use sensors like cameras, radar, and GPS.
- These sensors help the car see and understand its surroundings.
- The system allows the car to:
  - Detect objects
  - Make decisions
  - Control movement
- The SAE defines 6 levels of driving automation:
  - L0 = no automation
  - L5 = full self-driving

NAVIGATION TECHNIQUES AND CORE SENSORS

- LiDAR:
  - Creates detailed 3D maps using laser beams
  - Very accurate for shapes and distances
- Radar:
  - Uses radio waves to detect objects and speed
  - Less detailed than LiDAR and may miss small or non-metal objects
- Cameras:
  - Capture colors and images like human eyes
  - Used for reading lanes, signs, and object recognition

SENSOR FUSION CONCEPTS AND TECHNIQUES

- Sensor fusion means combining data from multiple sensors into one reliable result
- It improves accuracy, consistency, and reliability
- It is needed because:
  - Cameras fail in low light
  - Radar has low detail
  - LiDAR is affected by weather conditions
- Combining sensors helps reduce errors and improve safety

SENSOR FUSION CONCEPTS AND TECHNIQUES

- Kalman Filter:
  - Used to estimate position and track movement over time
- EKF & UKF:
  - Improved versions of Kalman Filter
  - Used for complex and non-linear movements
- Particle Filter:
  - Uses many possible states (particles)
  - Chooses the most likely correct position or result

HD MAPS AND LOCALIZATION

عبدالرحمن علاء احمد

HD MAPS AND LOCALIZATION

INTRODUCTION TO HD MAPS & LOCALIZATION

- Autonomous vehicles require high-precision positioning
- GNSS alone is not sufficient
- Solution:
  - HD Maps
  - Localization techniques

Goal:

- Achieve lane-level accuracy
- Ensure safe and reliable navigation

HD MAPS

What are HD Maps?

- High-Definition digital maps
- Provide centimeter-level accuracy
- Contain:
  - Lane boundaries
  - Traffic signs & signals
  - Road geometry

Key Feature:

- Rich environmental and semantic data

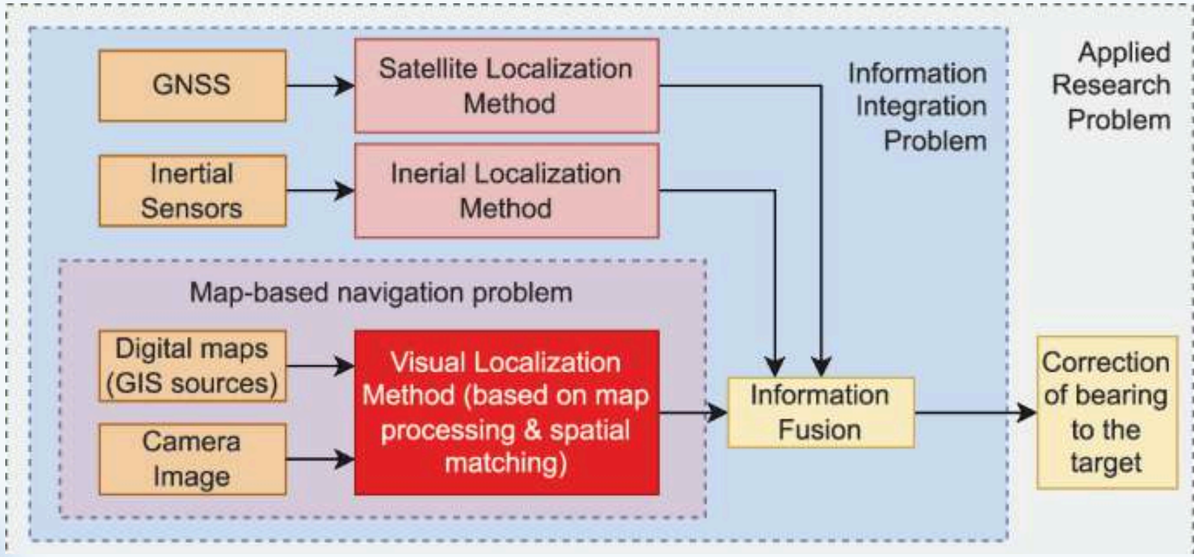
Localization Techniques

03

LOCALIZATION = DETERMINING VEHICLE POSITION ON MAP

TECHNIQUES:

- GNSS (GPS / RTK)
- LIDAR-BASED LOCALIZATION
- VISION-BASED LOCALIZATION
- SENSOR FUSION (KALMAN FILTER)



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Advantages & Limitations

ADVANTAGES:

- HIGH ACCURACY
- IMPROVED SAFETY
- BETTER DECISION-MAKING

LIMITATIONS:

- HIGH COST
- REQUIRES FREQUENT UPDATES
- LARGE DATA SIZE

ROLE IN AUTONOMOUS VEHICLES

- ENABLES LANE-LEVEL POSITIONING
- SUPPORTS PATH PLANNING
- **WORKS WITH:**
  - LIDAR
  - CAMERAS
  - INS

CONCLUSION:

HD MAPS + LOCALIZATION = RELIABLE AUTONOMOUS NAVIGATION

DIFFERENTIAL GPS AND REAL-TIME KINEMATIC

MOHAMED MOSTAFA

INTRODUCTION TO GPS AND PROBLEM

01

WHAT IS GPS?

Global Positioning System (GPS) is a satellite-based navigation system used to determine the exact location of a receiver anywhere on Earth. It works by receiving signals from multiple satellites and calculating position using trilateration.

Despite its wide use, standard GPS accuracy is limited (5–10 meters accuracy) due to several error sources that affect signal transmission.

01 **Atmospheric Errors**
Signal delay occurs when passing through the ionosphere and troposphere, affecting accuracy.

02 **Satellite Clock Errors**
Small inaccuracies in satellite clocks lead to incorrect time measurements and position errors.

03 **Multipath Effect**
Signals may reflect off buildings or surfaces before reaching the receiver, causing distortion.

DIFFERENTIAL GPS (DGPS)

02

Differential GPS (DGPS) is an enhancement technique used to improve the accuracy of standard GPS. It relies on a fixed reference station (Base Station) with a known position to calculate errors in GPS signals and transmit correction data to nearby users in real time.

DGPS uses pseudorange (code-based) measurements to estimate and correct positioning errors.

Base Station Concept: A stationary receiver at a known location calculates the difference between actual and measured position.

Error Correction: The base station sends correction signals to users to reduce positioning errors.

Improved Accuracy: DGPS improves accuracy from 5–10 meters to about 1–3 meters.

REAL-TIME KINEMATIC (RTK)

03

Real-Time Kinematic (RTK) is an advanced positioning technique that provides very high accuracy by using carrier phase measurements instead of code-based signals. It relies on a base station and a rover to achieve real-time error correction with centimeter-level precision.

RTK uses carrier phase measurements and resolves integer ambiguity to achieve highly precise positioning.

Base & Rover Concept: A base station and a moving receiver (rover) simultaneously receive satellite signals and compare measurements.

Carrier Phase Measurement: Uses the phase of the GPS signal carrier for much higher accuracy than code-based methods.

High Accuracy: RTK achieves accuracy up to 1–2 centimeters in real time.

COMPARISON AND APPLICATION

04

Differential GPS (DGPS) and (RTK) both use base station to correct satellite errors. DGPS uses code-based which suitable for general navigation and asset tracking. RTK uses carrier-phase signals and complex mathematical ambiguity resolution to achieve much higher, centimeter for precision applications.

Feature	Standalone GPS	Differential GPS (DGPS)	Real-Time Kinematic (RTK)
Accuracy	5–10 m	1–3 m	1–2 cm
Technique	Standard Satellite Signals	Code-based Corrections (Pseudorange)	Carrier Phase Measurements
System Setup	Receiver Only	Base Station + Receiver	Base Station + Rover + Continuous Connection
Applications	Consumer Navigation Driving Smartphone Apps	Asset Tracking GIS Mapping Rural Navigation	Precision Agriculture Land Surveying Autonomous Vehicles

Cybersecurity in Vehicle Navigation Systems

Mohamed Ghanem

The Navigation Attack Surface

Cybersecurity in Vehicle Navigation Systems

- **The Core Weakness**: Standard GPS/GNSS signals are incredibly weak and unencrypted, making them easy to manipulate.
- **A Tangled Web**: Modern vehicles don't just use GPS; they rely on a vulnerable network of sensors (cameras, LiDAR) and cloud connections (V2X).
- **The Stakes**: For autonomous cars, ships, and aircraft, a compromised navigation system isn't just an IT failure—it's an immediate safety threat to the vehicles

01

How Vehicles are Attacked

02

Signal & Sensor Vulnerabilities

- **GPS Spoofing**: Attackers broadcast powerful, fake satellite signals to secretly hijack a vehicle's location and trajectory.
- **Sensor Tampering**: Hackers can use targeted soundwaves to disrupt gyroscopes, or lasers to project "phantom obstacles" into a car's LiDAR.
- **Visual Deception**: AI perception systems can be easily tricked—for example, using carefully placed stickers to make a car read a stop sign as a speed limit sign.
  -

The Encryption Problem

Securing the Data Layer

- **The Unencrypted Threat**: Without strong encryption, hackers can intercept data to feed the vehicle fake map updates or roll back critical safety software.

- **The Speed Limit**: Vehicle-to-Vehicle (V2X) hazard alerts require ultra-fast verification (milliseconds). Heavy encryption algorithms can be too slow for real-time driving.

- **The Legacy Hurdle**: Cars stay on the road for 15-20 years. Retrofitting modern cryptographic security into older, embedded vehicle hardware is nearly impossible.

03

How to Fix It

04

Defense in Depth

- **Layered Security**: There is no "silver bullet." Defenses must be layered so that if a hacker breaches the GPS, the other sensors catch the error.

- **Cross-Validation**: Vehicles must constantly compare data. If the GPS says the car is turning, but the physical inertia sensors (INS) say it's going straight, the system flags an attack.

- **Secure OTA Updates**: Security isn't static. Vehicles need secure Over-The-Air updates to patch new vulnerabilities in real-time.

Conclusion

Security as a Core Requirement

- **Security is Not a Luxury:** As we enter the era of autonomous driving, navigation security is the first line of defense for human life.

- **Built-In, Not Added-On**: Cybersecurity can no longer be an afterthought; it must be engineered into vehicles from day one.

05

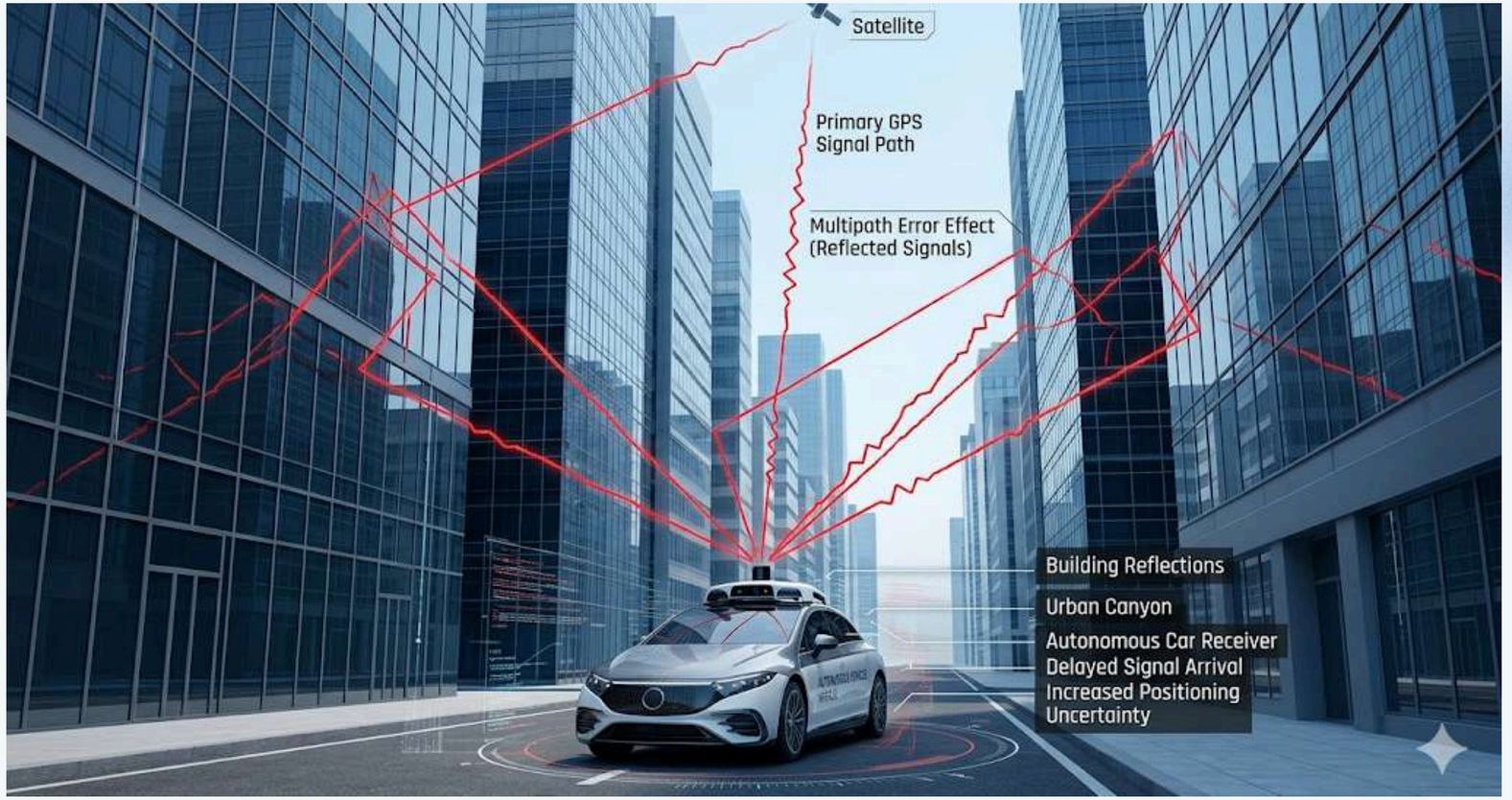
NAVIGATION IN URBAN VS RURAL ENVIRONMENTS

HOSSAM SALAH IBRAHIM

The Operational Design Domain (ODD): Navigation reliability varies drastically by environment.

Dynamic Sensor Trust Models: Autonomous vehicles cannot rely on a single fixed sensor setup.

The Shift: Transitioning between dense cities and open roads requires swapping priority between sensors.



URBAN ENVIRONMENTS

The Problem: "Urban Canyon" Effect.

- Tall skyscrapers block and reflect GNSS signals (Multipath errors).
- Highly dynamic elements (pedestrians, cyclists, complex intersections)

The Solution: Perception & HD Maps.

- Downgrade trust in GNSS.
- High trust in LiDAR, Visual SLAM, and semantic feature tracking (buildings, traffic lights)

RURAL ENVIRONMENTS

The Problem: Poor Perception Conditions.

- Lack of clear lane markings and street lighting.
- Unpaved roads, dust, or snow.
- Visual SLAM and LiDAR struggle to find trackable features.

The Solution: Absolute Positioning.

- Clear sky = Optimal GNSS & RTK performance.
- High trust in RTK-GNSS and INS.
- Cameras are shifted from localization to obstacle detection.

SUMMARY

The Inverted Trust Model:

- Urban: Low GNSS Trust >> High Perception Trust.
- Rural: High GNSS Trust >> Low Perception Trust (for localization).

Conclusion:

Seamless integration and dynamic adaptation are the keys to Level 5 Autonomy.

FUTURE TRENDS IN AUTONOMOUS NAVIGATION

YOUSSEF OSAMA ELSAYED

INTRODUCTION TO FUTURE TRENDS

Future Trends in Autonomous Navigation

- Autonomous vehicle navigation is evolving rapidly.
- Integration of GPS, IMU, LiDAR, Cameras, and Radar improved navigation.
- Future systems aim for Level 4 and Level 5 autonomy.
- Key trends include:
  - AI-driven perception
  - V2X communication
  - Quantum positioning
  - Neuromorphic computing
  - HD mapping and SLAM
- Goal: Safer, more accurate, and more reliable navigation systems

AI AND DEEP SENSOR FUSION

1 Deep learning improves environmental understanding

2 **CNNs and Vision Transformers enhance:**

- Object detection
- Lane detection
- Traffic sign recognition

3 **Early and mid sensor fusion combines:**

- LiDAR
- Cameras
- Radar

4 **Benefits**

- Better accuracy
- Robust performance in bad weather
- End-to-end decision making

5 **AI also provides uncertainty estimation for safer decisions**

V2X COMMUNICATION AND POSITIONING

V2X Communication

- Vehicle to Vehicle (V2V)
- Vehicle to Infrastructure (V2I)
- Vehicle to Pedestrian (V2P)
- Vehicle to Network (V2N)

Benefits:

- Cooperative navigation
- Real-time data sharing
- Better urban positioning
- Smart infrastructure support

Next-Generation GNSS

- Multi-constellation systems:
  - GPS
  - Galileo
  - BeiDou
  - GLONASS
- PPP and RTK provide centimeter-level accuracy
- Quantum sensors may enable GNSS-independent positioning

FUTURE TECHNOLOGIES

- Neuromorphic computing for energy-efficient AI
- Edge computing for real-time processing
- HD Maps and Deep SLAM
- Multi-modal sensors
- Cybersecurity protection

Challenges

- Rare edge-case scenarios
- System validation
- Ethical and regulatory issues
- Infrastructure requirements

Conclusion

Future autonomous navigation will rely on:

AI + V2X + GNSS + Quantum + Smart Infrastructure = Intelligent Mobility

THANK YOU